Testing on live application

SESSION-4- API Testing (Basic Intro) + Database SQL Testing

1.Use the online API testing tool ReqRes (https://reqres.in/) to make a GET request to /api/users/2 and note down the response status code and the returned user's first name.

Ans: ReqRes API Testing – GET Request

- A GET request was sent to the /api/users/2 endpoint using the ReqRes API testing tool.
- The server processed the request successfully and returned the response status code 200 OK.
- The response body was received in JSON format.
- The user information was available inside the data object.
- The returned user's first name value was "Janet".
- Other user details such as id, email, last name, and avatar URL were also included in the response.
- No error messages, delays, or malformed data were observed.
- The API response structure matched the expected format defined in the documentation.
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2.Send a POST request using ReqRes to /api/users with JSON data {"name": "Virat", "job": "Cricketer"} and write down the response code and the id value returned in the response.

Ans: ReqRes API Testing – POST Request

Create a new user using a POST request and verify the response.

{
"name": "Virat",
"job": "Cricketer"
}

- A POST request was sent to the /api/users' endpoint with valid JSON data.
- The request payload contained two fields: name and job.
- The server accepted the request and returned the response status code 201 Created.
- The response body included the submitted values along with additional fields generated by the server.
- A unique id value was returned in the response.
- A created at timestamp was also generated automatically.
- The generated id value changed every time the request was executed, indicating dynamic record creation.
- The response data matched the submitted request payload, confirming successful API processing.

3.Given the following JSON response from an API: {"id": 101, "name": "Pizza Hut", "location": "Ahmedabad"}, write the SQL INSERT statement to add this restaurant to a table named restaurants with columns id, name, and location.

Ans:

Given JSON Response:

{
"id": 101,
"name": "Pizza Hut",
"location": "Ahmedabad"
}

SQL Query:

```
INSERT INTO restaurants (id, name, location)
VALUES (101, 'Pizza Hut', 'Ahmedabad');
```

- The INSERT statement adds a new record to the restaurants table.
- The values from the JSON response are mapped directly to the corresponding database columns.
- The id field is stored as an integer value.
- The name and location fields are stored as string values enclosed within single quotes.
- After successful execution, one new row should be added to the database.
- The inserted record can be verified using a SELECT query.

4.Suppose you have a users table in your database and the UI shows a new user 'Riya' was registered. Write an SQL SELECT query to confirm if 'Riya' exists in the users table, and explain what result you would expect if the UI and DB are in sync.

Ans: SQL Query to Verify User Registration

SQL Query:

```
SELECT *
FROM users
WHERE name = 'Riya';
```

- This query searches the users table for records where the name column contains the value, Riya.

- If the registration process completed successfully, the query should return at least one matching row.
- The returned data may include user information such as ID, email address, phone number, and registration timestamp.
- If the query returns no rows, it indicates that the user registration data was not saved correctly.
- Successful synchronization between the UI and database confirms that the backend functionality is working as expected.
- This validation helps testers ensure data consistency between the front-end application and the database.

**5.After deleting a product from your shopping cart in the Flipkart UI, write an SQL query to verify that the product is no longer present in the cart_items table for your user.

Hint: Use SELECT to check for the product_id and your user_id in cart_items. **

Ans: SQL Query to Verify Product Deletion from Shopping Cart

Verify that a deleted product is no longer present in the database.

SQL Query:

```
SELECT *
FROM cart_items
WHERE user_id = 101
AND product_id = 401;
```

- This query checks whether the specified product still exists in the user's cart.
- The user_id identifies the logged-in user, while the product_id identifies the deleted product.
- After removing the product from the Flipkart UI, the query should return zero rows.

- If any record is returned, it indicates that the product was not deleted successfully from the database.
- This verification ensures that the UI action is correctly synchronized with the backend database.
- Testers should replace the sample user_id and product_id values with actual test data.
- Database validation is important for confirming data integrity and preventing inconsistencies between the UI and backend systems.